Realised volatility, heterogeneous autoregressive (HAR) modelling and a VIX-based trading strategy

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Abstract

This study analyses the dynamics of realised volatility for the S&P 500 index using a heterogeneous autoregressive (HAR) specification. Starting from daily index data, we construct monthly realised volatility at 1-, 3- and 6-month horizons and regress next-month volatility on these components. The one-month term is the main driver of short-horizon dynamics; medium- and long-run components add little incremental explanatory power once short-run volatility is controlled for. The results are robust to heteroskedasticity- and autocorrelation-consistent inference and to an out-of-sample forecasting exercise over 2018–2025. Finally, we translate the HAR forecasts into trading signals by comparing forecast realised volatility with option-implied volatility (VIX) and feeding the resulting spread into a simple rule-based trading bot that takes long/short positions in a short-term VIX futures ETF. The strategy delivers economically meaningful performance but entails sizeable drawdowns, illustrating both the potential and the limits of exploiting HAR-based volatility forecasts in practice.

1 Introduction

Volatility forecasting is central to risk management, derivative pricing and portfolio allocation. A large literature documents that volatility is highly persistent and exhibits heterogeneous dynamics across multiple time horizons. Within this literature, the heterogeneous autoregressive (HAR) model of realised volatility proposed by Corsi (2009) has become a widely used benchmark. By combining volatility components measured over different horizons, the HAR model provides a simple and interpretable approximation to long-memory behaviour in volatility.

The objective of this work is to implement a HAR model for realised volatility of the S&P 500 index and to assess the role of medium- and long-term volatility components in explaining next-month volatility. We focus on realised volatility measured over 1-, 3- and 6-month horizons and evaluate their relative contribution both in-sample and in an out-of-sample forecasting exercise. In a second step, we compare HAR-based forecasts of future realised volatility with option-implied volatility extracted from the VIX index and use the resulting forecast–implied spread to generate trading signals for a simple rule-based volatility trading bot.

The remainder of the paper is organised as follows. Section 2 describes the data and sampling scheme. Section 3 details the construction of realised volatility measures. Section 4 presents preliminary time-series evidence. Section 5 sets out the HAR specification and discusses estimation results, including HAC-robust inference and residual diagnostics. Section 6 evaluates out-of-sample forecast performance. Section 7 introduces a VIX-based volatility trading bot built on the HAR forecasts. Section 8 concludes.

2 Data and sampling scheme

2.1 Underlying price data

The empirical analysis is based on daily observations of the S&P 500 index over an extended sample period. Let P_t denote the closing level of the index on trading day t. Daily log-returns are defined as

$$r_t = \log P_t - \log P_{t-1},$$

and constitute the primitive input for constructing realised measures of variance and volatility. Throughout, all realised volatility measures are computed from these log-returns.

2.2 From daily data to a monthly panel

The HAR framework is formulated in terms of volatility components measured over different horizons but is typically estimated on a lower-frequency panel. In our setting, the target frequency is approximately monthly. To this end, we construct a sequence of equally spaced observation dates by retaining one day every 22 trading days from the original daily series. That is, starting from the full daily sample, we keep every 22nd trading day and discard intermediate dates.

This procedure delivers a panel indexed by business time, where each observation is separated from the next by a fixed number of trading days. It yields intervals of roughly one calendar month while remaining independent of calendar-month boundaries and specific calendar effects. In particular, the sampling scheme avoids systematic alignment with end-of-month or day-of-week patterns.

At each selected date t, monthly realised variance, denoted RVarM_t, is computed as the average of daily realised variances over the preceding 22 trading days. Realised volatility at the 1-, 3- and 6-month horizons (RVol1M, RVol3M and RVol6M) is then obtained by aggregating returns over the previous 22, 66 and 132 trading days, respectively, and taking

the square root of the corresponding realised variances.

2.3 Monthly panel dataset

The resulting monthly panel dataset (Figure 1) contains, for each retained date t:

- the S&P 500 index level (SP_t);
- realised variance at the 1-month horizon (RVarM_t);
- realised volatility at the 1-, 3- and 6-month horizons ($\mathrm{RVol1M}_t$, $\mathrm{RVol3M}_t$, $\mathrm{RVol6M}_t$);
- next-month realised volatility ($\mathrm{RVolM{+}1}_t$), computed over the 22 trading days following date t.

	Date	SP	RVarM	RVolM	RVol3M	RVol6M	RVolM+1
5	1990-07-11	361.230011	0.016989	0.130342	0.120901	0.127600	0.171046
6	1990-08-10	335.519989	0.029257	0.171046	0.144205	0.127298	0.225798
7	1990-09-12	322.540009	0.050985	0.225798	0.175729	0.144842	0.219850
8	1990-10-12	300.029999	0.048334	0.219850	0.205565	0.163233	0.202199
9	1990-11-13	317.670013	0.040884	0.202199	0.215949	0.180077	0.121797
...	...	...	...	...	...	...	...
324	2018-05-17	2720.129883	0.010931	0.104552	0.157823	0.140828	0.089645
325	2018-06-19	2762.590088	0.008036	0.089645	0.139881	0.144958	0.095119
326	2018-07-20	2801.830078	0.009048	0.095119	0.096439	0.149593	0.078383
327	2018-08-21	2862.959961	0.006144	0.078383	0.087716	0.122769	0.063200
328	2018-09-21	2929.669922	0.003994	0.063200	0.078901	0.109391	NaN

Figure 1: Structure of the monthly panel with S&P 500 index levels, realised variance and realised volatilities.

This monthly panel constitutes the input for all subsequent econometric analyses, with $\mathrm{RVolM{+}1}_t$ serving as the dependent variable in the HAR regressions and the lagged realised volatility components as regressors.

3 Construction of realised volatility measures

3.1 Daily realised variance and volatility

In the realised-volatility literature, ex-post variance over a trading day is defined as the sum of squared intraday returns. Let $r_{t,i}$ denote the i-th intraday return on day t, for $i = 1, \ldots, M$. The daily realised variance is then

$$\mathrm{RVarD}_t = \sum_{i=1}^{M} r_{t,i}^2. \tag{1}$$

High-frequency returns are not available in our setting. Following common practice when only daily closing prices are observed, we approximate RVarD_t using the squared daily log-return, so that

$$\mathrm{RVarD}_t \approx r_t^2,$$

and define daily realised volatility as

$$\mathrm{RVolD}_t = \sqrt{\mathrm{RVarD}_t}. \tag{2}$$

This low-frequency proxy does not exploit intraday information but provides a consistent measure of ex-post variation at the daily horizon.

3.2 One-month realised variance and volatility

To obtain a monthly measure, daily realised variance is aggregated over a rolling window of 22 trading days, in line with the HAR-RV framework of Corsi (2009). The one-month realised variance at date t is defined as

$$\mathrm{RVar}_t^{(1m)} = \frac{1}{22} \sum_{j=0}^{21} \mathrm{RVarD}_{t-j}, \tag{3}$$

and the corresponding one-month realised volatility as

$$\mathrm{RVol}_t^{(1m)} = \sqrt{\mathrm{RVar}_t^{(1m)}}. \tag{4}$$

In the empirical analysis, this one-month realised variance and volatility are stored as `RVarM` and `RVol1M` in the monthly panel introduced above. Any subsequent annualisation is obtained by the usual square-root-of-time scaling and does not affect the comparative properties of the volatility components.

3.3 Three- and six-month components

The HAR model emphasises volatility components measured over different horizons. In our specification, we construct medium- and long-term components by smoothing the one-month realised volatility series over 3- and 6-month windows. Specifically,

$$\mathrm{RVol}_t^{(3m)} = \frac{1}{3}\sum_{j=0}^{2} \mathrm{RVol}_{t-j}^{(1m)}, \tag{5}$$

$$\mathrm{RVol}_t^{(6m)} = \frac{1}{6}\sum_{j=0}^{5} \mathrm{RVol}_{t-j}^{(1m)}. \tag{6}$$

These series, denoted `RVol3M` and `RVol6M` in the dataset, capture medium- and long-horizon volatility dynamics and serve as regressors in the HAR specification alongside the one-month component `RVol1M`.

Finally, to obtain a forecasting set-up consistent with the one-step-ahead structure of Corsi (2009), we align all regressors so that volatility over the interval preceding date t is used to predict volatility over the subsequent month. Let

$$\mathrm{RVolM{+}1}_t = \mathrm{RVol}_{t+1}^{(1m)}$$

denote realised volatility over the month following date t; this variable, stored as `RVolM+1`

in the dataframe, is the dependent variable in the HAR regressions.

4 Preliminary time-series evidence

4.1 Persistence of monthly realised volatility

We begin by examining the time-series behaviour of the monthly realised volatility series used in the HAR specification. Both the contemporaneous one-month component, RVol1M_t, and the forward-looking target, RVolM+1_t, display the familiar pattern of volatility clustering, with periods of relatively tranquil markets interspersed with episodes of markedly elevated volatility that tend to persist over several months. These features are consistent with the stylised facts documented for equity index volatility and already suggest a highly persistent data-generating process.

4.2 Autocorrelation structure

To obtain a more formal description of persistence, we compute the sample autocorrelation function (ACF) and partial autocorrelation function (PACF) of monthly realised volatility. Figure 2 reports the ACF and PACF for RVolM+1_t over a range of lags.

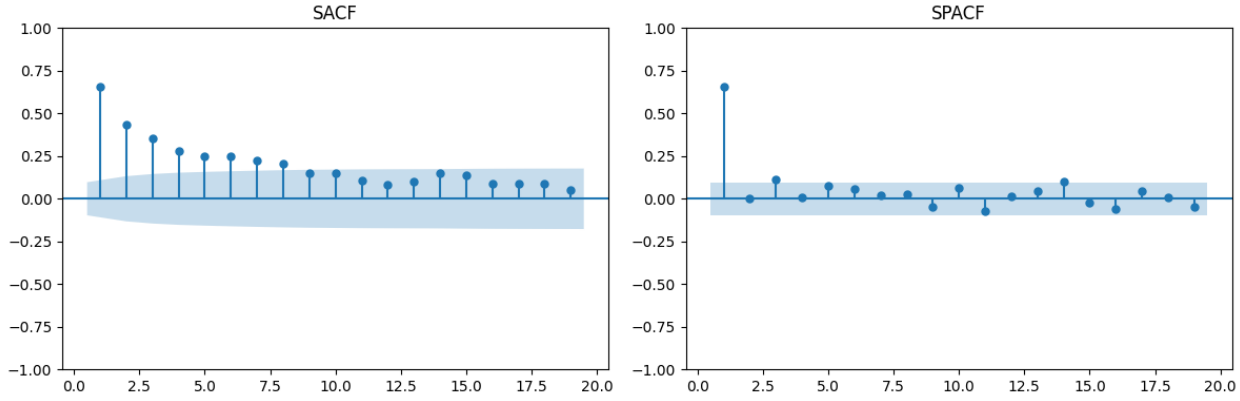

Figure 2: Sample autocorrelation function (left) and partial autocorrelation function (right) of monthly realised volatility.

The ACF displays a very pronounced first-order autocorrelation, with the lag-1 coefficient

around 0.6–0.7. Autocorrelations then decline monotonically as the lag length increases and become small and mostly insignificant after a moderate number of lags. This pattern is characteristic of a highly persistent but ultimately short-memory process whose dependence decays at an approximately geometric rate.

The PACF is dominated by a single large spike at lag 1, while the remaining coefficients lie within, or only marginally outside, the confidence bands. The absence of sizeable spikes at higher orders is consistent with a low-order autoregressive representation, in particular an AR(1) specification, rather than a process that requires many lags to capture its dynamics.

Taken together, the ACF and PACF suggest that monthly realised volatility is strongly persistent but can be well approximated by a relatively parsimonious dynamic model with only a small number of autoregressive terms.

4.3 Implications for model specification

The preliminary evidence therefore motivates a specification that (i) recognises the strong persistence of realised volatility and (ii) allows different segments of the past volatility history to exert distinct effects on current volatility. The HAR model achieves this by including one-month, three-month and six-month realised volatility components as separate regressors. Although the sample ACF and PACF do not point to genuine long-memory behaviour, the HAR structure offers a flexible yet low-dimensional way of capturing the observed persistence by combining short-, medium- and longer-horizon volatility components. In the subsequent section, we adopt this structure and examine the relative contribution of these components to the prediction of next-month realised volatility RVolM+1_t.

5 Econometric framework: HAR specification

5.1 Model specification

Let $\mathrm{RVol}_t^{(1m)}$ denote one-month realised volatility at time t, and let $\mathrm{RVol}_t^{(3m)}$ and $\mathrm{RVol}_t^{(6m)}$ be the corresponding 3- and 6-month components constructed as in the previous section. The forecasting target is next-month realised volatility, denoted $\mathrm{RVolM{+}1}_t$, i.e. realised volatility over the 22 trading days following date t.

In line with the heterogeneous autoregressive (HAR-RV) model of Corsi (2009), we adopt a linear specification in which volatility components measured over different horizons enter as separate regressors. In our monthly setting, the baseline HAR model is

$$\mathrm{RVolM{+}1}_t = \alpha + \beta_1 \,\mathrm{RVol}_t^{(1m)} + \beta_3 \,\mathrm{RVol}_t^{(3m)} + \beta_6 \,\mathrm{RVol}_t^{(6m)} + \varepsilon_t, \tag{7}$$

where ε_t is an innovation term with zero mean and finite variance. The one-month component $\mathrm{RVol}_t^{(1m)}$ captures short-run volatility persistence at the horizon relevant for the forecast, whereas $\mathrm{RVol}_t^{(3m)}$ and $\mathrm{RVol}_t^{(6m)}$ proxy for medium- and long-run volatility trends. Equation (7) is thus the direct monthly analogue of the standard HAR-RV model, which typically combines daily, weekly and monthly realised volatility components.

5.2 Estimation and inference

The model in (7) is estimated by ordinary least squares (OLS) on the monthly panel described above. After dropping the last observation to align regressors and dependent variable, the estimation sample comprises 315 monthly observations. Consistently with the related literature, we work with realised volatility levels rather than with logarithms, so that coefficients can be interpreted directly in terms of marginal effects on the conditional mean of volatility.

The OLS estimation delivers an R^2 of about 0.55, indicating that the three volatility components jointly explain a substantial share of the variation in next-month realised volatility

despite the relatively coarse (monthly) sampling. Semi-partial R^2 measures reveal that almost all of this explanatory power is concentrated in the one-month component $\mathrm{RVol}_t^{(1m)}$, whereas the incremental contribution of the 3- and 6-month components is modest. This pattern anticipates the empirical results reported below.

Because realised volatility series are typically affected by conditional heteroskedasticity and mild serial correlation in the regression residuals, we also re-estimate (7) using heteroskedasticity- and autocorrelation-consistent (HAC) standard errors of the Newey–West type. In our application, the use of HAC standard errors increases the estimated standard errors of all coefficients and raises the overall F-statistic, but does not alter the qualitative conclusions: the coefficient on $\mathrm{RVol}_t^{(1m)}$ remains strongly statistically significant, while the evidence in favour of distinct 3- and 6-month effects remains comparatively weak.

5.3 Theoretical background and expectations

From a theoretical standpoint, the specification in (7) is motivated by the heterogeneous market hypothesis and the associated volatility cascade stressed by Corsi (2009). Different groups of market participants operate at different horizons, and their trading decisions generate volatility components that propagate from longer to shorter horizons. The HAR-RV model provides a simple AR-type representation of this volatility cascade by including distinct volatility aggregates at multiple horizons.

In such a framework, we expect the one-month component to exert the largest influence on next-month volatility, reflecting strong short-run persistence and slow mean reversion. Medium- and long-horizon components should retain some positive predictive content, as they capture low-frequency shifts in the volatility level, but their marginal contribution is expected to decline with the horizon and to become small for components approaching the annual scale. The empirical analysis that follows is designed precisely to quantify the relative importance of the 1-, 3- and 6-month components and to assess whether the longer-run terms add economically meaningful predictive power once short-run volatility is controlled for.

6 OLS estimation with standard covariance

6.1 Model fit

The HAR regression in (7) is first estimated by ordinary least squares using the conventional (non-robust) covariance matrix. The dependent variable, denoted `RVolM+1` in the dataset, is next-month realised volatility, while the regressors are the 1-, 3- and 6-month realised volatility components (`RVol1M`, `RVol3M` and `RVol6M`).

The regression is estimated on 315 monthly observations. The model attains an R^2 of 0.547 and an adjusted R^2 of 0.543. The F-statistic for the joint significance of the three volatility regressors equals 125.3 with an associated p-value of 3.18×10^{-53}. Thus, the HAR specification explains slightly more than half of the total variation in next-month realised volatility and the regressors are jointly highly significant in the usual OLS sense.

6.2 Coefficient estimates

Under the standard covariance matrix, the estimated coefficients are:

- constant: $\hat{\alpha} = 0.0297$ (standard error 0.008, $t = 3.666$, $p < 0.001$; 95% confidence interval [0.014, 0.046]);
- one-month volatility: $\widehat{\beta}_1 = 0.6240$ (standard error 0.077, $t = 8.102$, $p < 0.001$; 95% confidence interval [0.472, 0.775]);
- three-month volatility: $\widehat{\beta}_3 = 0.0295$ (standard error 0.138, $t = 0.214$, $p = 0.830$; 95% confidence interval [−0.242, 0.301]);
- six-month volatility: $\widehat{\beta}_6 = 0.1489$ (standard error 0.115, $t = 1.291$, $p = 0.198$; 95% confidence interval [−0.078, 0.376]).

OLS Regression Results

Dep. Variable:	RVolM+1	**R-squared:**	0.536
Model:	OLS	**Adj. R-squared:**	0.531
Method:	Least Squares	**F-statistic:**	122.7
Date:	Sun, 30 Nov 2025	**Prob (F-statistic):**	7.62e-53
Time:	21:00:34	**Log-Likelihood:**	441.46
No. Observations:	323	**AIC:**	-874.9
Df Residuals:	319	**BIC:**	-859.8
Df Model:	3		
Covariance Type:	nonrobust		

	coef	**std err**	**t**	**P>|t|**	**[0.025**	**0.975]**
const	0.0308	0.008	3.810	0.000	0.015	0.047
RVolM	0.6162	0.076	8.055	0.000	0.466	0.767
RVol3M	0.0336	0.137	0.246	0.806	-0.236	0.303
RVol6M	0.1451	0.114	1.270	0.205	-0.080	0.370

Omnibus:	176.662	**Durbin-Watson:**	1.966
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	1820.337

Figure 3: OLS regression output with non-robust covariance matrix.

The one-month coefficient is positive, large in magnitude and strongly statistically significant. Its estimated value of around 0.62 implies that, ceteris paribus, a one-point increase in the one-month realised volatility component is associated with an increase of roughly 0.6 points in expected next-month volatility. By contrast, the coefficients on the 3- and 6-month components are imprecisely estimated and statistically insignificant at conventional levels, even though the point estimate on the 6-month component is positive and of non-negligible size.

6.3 Semi-partial R^2 and incremental contributions

To quantify the marginal contribution of each horizon, we compute semi-partial R^2 statistics. For a given regressor, the semi-partial R^2 is defined as the increase in the overall R^2 that arises when that regressor is added to a model that already contains all other regressors.

The estimated semi-partial R^2 values for the three volatility components are:

- one-month realised volatility: 0.094447;

- three-month realised volatility: 0.000088;
- six-month realised volatility: 0.002349.

	Monthly Semi-Partial R2	Trimester Semi-Partial R2	Semester Semi-Partial R2
0	0.094447	0.000088	0.002349

Figure 4: Semi-partial R^2 for the 1-, 3- and 6-month realised volatility components.

The one-month component alone accounts for an additional 9.4 percentage points of explained variance in next-month realised volatility, over and above what is captured by the other regressors. In contrast, the 3- and 6-month components contribute virtually nothing in marginal terms: their semi-partial R^2 values are very close to zero. This confirms the dominant role of the 1-month horizon in the HAR representation and suggests that most of the predictive content of longer horizons is already embedded in the short-run component.

6.4 In-sample fit

Figure 5 compares realised volatility with the fitted values from the HAR regression over the estimation sample.

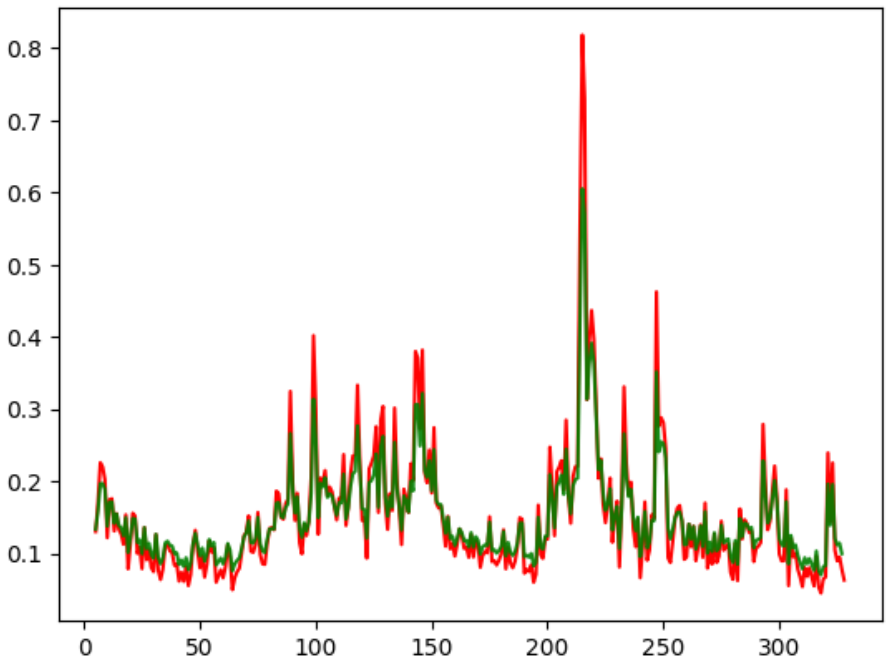

Figure 5: In-sample comparison between realised volatility and HAR fitted values.

The fitted series tracks the level and medium-run swings of realised volatility reasonably well, including large spikes followed by gradual mean reversion. The most visible discrepancies are concentrated in a limited number of extreme observations, which is consistent with the presence of heavy-tailed shocks in the volatility process.

7 HAC-robust inference

7.1 Motivation

Realised volatility series are typically characterised by conditional heteroskedasticity and mild serial correlation. Under these conditions, the classical OLS assumption of i.i.d. residuals is violated, and standard errors based on the non-robust covariance matrix may be misleading. To assess the robustness of the inference, we therefore re-estimate the same HAR regression using heteroskedasticity- and autocorrelation-consistent (HAC) standard errors in the spirit of Newey and West (1987).

7.2 Effect on standard errors and test statistics

HAC-robust estimation leaves the coefficient point estimates and overall fit measures (R^2, adjusted R^2, log-likelihood, AIC and BIC) unchanged. Only the estimated covariance matrix, and therefore standard errors and test statistics, is modified.

Under HAC standard errors, the key results can be summarised as follows:

- the constant remains at 0.0297, with a standard error of about 0.007 and a t-statistic of 4.601;
- the one-month coefficient remains 0.6240, but its standard error rises to approximately 0.141, lowering the t-statistic to 4.364; the 95% confidence interval widens accordingly (roughly [0.338, 0.894]);
- the three-month coefficient stays at 0.0295, with a slightly larger standard error (around 0.150), $t \approx 0.224$ and $p \approx 0.82$;
- the six-month coefficient remains 0.1489, with a standard error of about 0.082, yielding $t \approx 1.776$ and $p \approx 0.08$.

The F-statistic for joint significance increases from 125.3 to roughly 174.1, with an associated p-value of about 7.56×10^{-67}, confirming extremely strong joint significance even under the more conservative covariance specification.

OLS Regression Results

Dep. Variable:	RVolM+1	**R-squared:**	0.536
Model:	OLS	**Adj. R-squared:**	0.531
Method:	Least Squares	**F-statistic:**	174.1
Date:	Sun, 30 Nov 2025	**Prob (F-statistic):**	7.56e-67
Time:	21:00:34	**Log-Likelihood:**	441.46
No. Observations:	323	**AIC:**	-874.9
Df Residuals:	319	**BIC:**	-859.8
Df Model:	3		
Covariance Type:	HAC		

	coef	**std err**	**t**	**P>|t|**	**[0.025**	**0.975]**
const	0.0308	0.007	4.601	0.000	0.018	0.044
RVolM	0.6162	0.141	4.364	0.000	0.338	0.894
RVol3M	0.0336	0.150	0.224	0.823	-0.262	0.329
RVol6M	0.1451	0.082	1.776	0.077	-0.016	0.306

Omnibus:	176.662	**Durbin-Watson:**	1.966
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	1820.337

Figure 6: OLS regression output with HAC covariance matrix.

7.3 Interpretation

The HAC adjustment substantially reduces the apparent precision of the one-month coefficient relative to the non-robust case, as reflected in the larger standard error and lower t-statistic. Nonetheless, the coefficient on the 1-month component remains highly significant at any conventional level, and its magnitude is essentially unchanged. The three-month component remains clearly insignificant. The six-month coefficient moves closer to borderline significance at the 10% level, but its contribution remains statistically fragile and modest in economic terms.

Taken together with the semi-partial R^2 results, the HAC-based inference confirms that:

(i) the one-month realised volatility component is a robust and quantitatively important predictor of next-month volatility;

(ii) the three-month component offers no meaningful incremental information beyond what is already captured by the 1- and 6-month components;

(iii) the six-month component may reflect some slow-moving volatility trends, but its marginal contribution is small and only weakly supported by the data.

8 Residual diagnostics

8.1 Distributional properties and normality tests

The OLS output reports an Omnibus statistic of 174.654 and a Jarque–Bera statistic of 1908.506, both with p-values effectively equal to zero. These tests decisively reject the null hypothesis of normally distributed residuals. The residual skewness and kurtosis are equal to 2.042 and 14.346, respectively, indicating pronounced right-skewness and very heavy tails relative to the Gaussian benchmark.

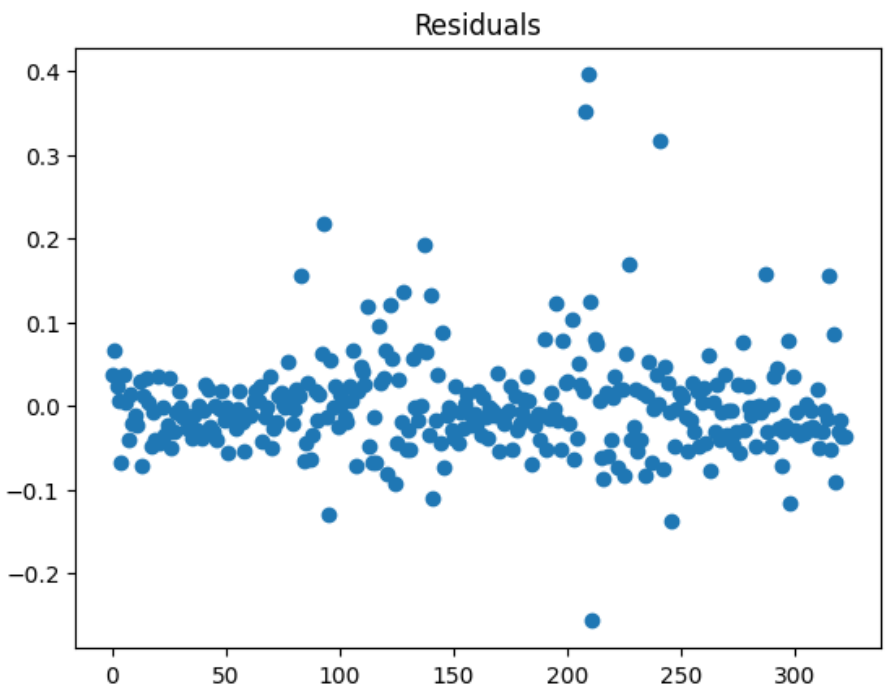

Figure 7: Residuals from the HAR regression.

As shown in Figure 7, the residuals are centred around zero and do not exhibit an obvious deterministic pattern, but there is clear clustering of large absolute values, particularly during periods of elevated market volatility. This pattern is consistent with volatility clustering in

the underlying process and with the heavy-tailed nature of volatility shocks highlighted by the formal normality tests.

8.2 Serial correlation and Durbin–Watson statistic

The Durbin–Watson statistic, equal to 1.949, is close to the benchmark value of 2, suggesting limited first-order autocorrelation in the residuals. This is consistent with the visual evidence from the residual plot, which does not reveal strong systematic serial dependence. However, the combination of heavy tails, conditional heteroskedasticity and potential higher-order dependence justifies the use of HAC-robust standard errors in the main inference, as discussed above.

Overall, the residual diagnostics confirm that the linear HAR specification captures the bulk of the predictable variation in next-month realised volatility, while leaving a residual component that is non-Gaussian and conditionally heteroskedastic—features that are typical of volatility time series.

9 Out-of-sample forecast performance

9.1 One-step-ahead forecasts, 2019–2025

To evaluate the model in a genuinely predictive setting, we conduct an out-of-sample forecasting exercise. Starting from an initial estimation window that covers the pre-2019 sample, the HAR model in (7) is re-estimated recursively: at each month t in the evaluation period, the parameters are estimated using only information available up to $t-1$, and a one-step-ahead forecast of next-month realised volatility, $\widehat{\mathrm{RVolM}+1}_t$, is produced. This real-time forecasting procedure is implemented over the period 2019–2025.

Figure 8 plots, over the out-of-sample window, the realised volatility series, the corresponding HAR forecasts and the VIX index as an option-implied measure of market volatility.

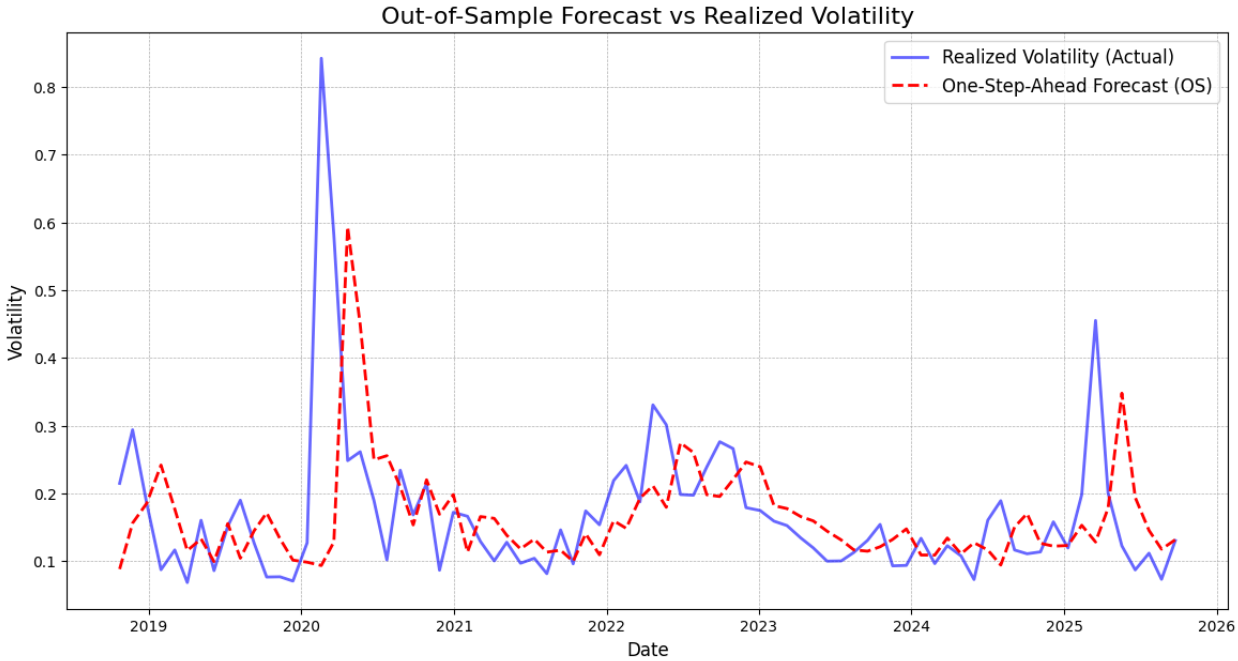

Figure 8: Out-of-sample one-step-ahead HAR forecasts, realised volatility and VIX, 2019–2025.

The HAR forecasts capture the broad swings in realised volatility, including the sharp spike associated with the onset of the COVID-19 crisis and the subsequent episodes of heightened market turbulence. As expected for a linear prediction model, extreme realised values are partially smoothed, but the timing and relative magnitude of most volatility surges are reproduced reasonably well. The comparison with the VIX index highlights the different nature of the two measures: VIX tends to react more abruptly to news and often anticipates realised volatility peaks, whereas the HAR forecasts reflect a smoothed and backward-looking assessment based on past realised volatility.

Standard forecast accuracy measures (such as mean squared error and mean absolute error, not reported here in detail) indicate that the HAR specification delivers non-trivial predictive content and is competitive with simple benchmarks based on constant or historical-average volatility. Overall, the out-of-sample evidence confirms that the predictive structure identified in-sample is not purely artefactual and that the one-month realised volatility component retains substantial forecasting power when evaluated in real time.

10 VIX-based volatility trading bot

10.1 Data, forecasts and spread

We finally investigate whether the HAR-based forecasts of realised volatility contain exploitable information for volatility trading. To this end, we construct a simple volatility-arbitrage trading bot that trades a listed volatility product based on the discrepancy between forecast realised volatility and option-implied volatility from VIX.

Daily data for the CBOE VIX index and a short-term VIX futures exchange-traded fund (ETF) with ticker `VIXY` are obtained from Yahoo Finance via the `yfinance` API. For each trading day, we retain the VIX closing level, denoted `VIX_Close`, and the ETF closing price, denoted `ETF_Close`. Daily ETF returns are computed as

$$\text{ETF_Return}_t = \frac{\text{ETF_Close}_t - \text{ETF_Close}_{t-1}}{\text{ETF_Close}_{t-1}}.$$

The monthly HAR estimation and forecasting exercise yields a time series of one-step-ahead forecasts of next-month realised volatility, stored in the variable `Forecasted_Rvolt+1`. These forecasts are first rescaled to be on the same numerical scale as VIX and then merged onto the daily VIX/ETF dataset by calendar date. Missing values are forward filled so that each forecast is held constant until the release of the next forecast. The resulting merged dataset contains, for each trading day, the VIX level, ETF return and the most recent HAR-based volatility forecast.

For each day t we define the forecast–implied spread

$$\text{Spread}_t = \text{Forecasted_Rvolt+1}_t - \text{VIX_Close}_t,$$

which measures the difference between model-implied future realised volatility and contemporaneous option-implied volatility. A positive spread indicates that future realised volatility is expected to exceed implied volatility (volatility underpriced), while a negative spread sug-

gests the opposite.

10.2 Signal rule and parameter sweep

Trading signals are generated by a threshold rule applied to the spread. Let $\theta \geq 0$ denote a mispricing threshold. The signal generator `VolForecastSignal` implements the following decision rule:

- if $\text{Spread}_t > \theta$, take a long position in volatility (long `VIXY`);
- if $\text{Spread}_t < -\theta$, take a short position in volatility (short `VIXY`);
- if $|\text{Spread}_t| < \theta$, stay flat.

Formally, an intermediate signal is constructed as $\text{sig}_t = \text{sign}(\text{Spread}_t)$, which is then set to zero whenever $|\text{Spread}_t| < \theta$. To avoid look-ahead bias, the resulting signal is shifted by one day, so that the position applied on day t is based solely on information available at the close of day $t-1$.

We consider a grid of five threshold values,

$$\theta \in \{0.0,\ 1.0,\ 2.0,\ 3.0,\ 5.0\},$$

which defines five versions of the trading bot. The case $\theta = 0$ corresponds to a pure sign rule that always takes a non-zero position whenever the forecast and VIX differ, while larger values of θ require more pronounced forecast–implied discrepancies before entering a trade.

10.3 Backtesting methodology

The backtesting engine `VolArbBacktester` takes as input a signal generator and the merged VIX/ETF/forecast dataset. For a given threshold θ, the daily position series $\text{Position}_t \in$

$\{-1, 0, 1\}$ is constructed as described above. The daily strategy return is then

$$\text{Strategy_Return}_t = \text{Position}_t \times \text{ETF_Return}_t,$$

so that the strategy is long (short) volatility when the signal is positive (negative), and flat otherwise. Starting from an initial capital $\text{Equity}_0 = 100{,}000$, the equity curve is obtained by compounding returns:

$$\text{Equity}_t = \text{Equity}_0 \prod_{s \le t} (1 + \text{Strategy_Return}_s).$$

For each threshold value we compute three performance measures:

- the total gross return,

$$\text{TotalReturn} = \frac{\text{Equity}_T}{\text{Equity}_0} - 1,$$

  over the available backtest window;

- the annualised Sharpe ratio, based on daily equity returns and a 252-day trading year;

- the maximum drawdown,

$$\text{MaxDD} = \max_t \left(\frac{\max_{s \le t} \text{Equity}_s - \text{Equity}_t}{\max_{s \le t} \text{Equity}_s} \right),$$

  capturing the largest peak-to-trough decline in the equity curve.

Transaction costs, funding costs and leverage constraints are ignored, so all results should be interpreted as gross and potentially optimistic.

10.4 Results

Table 1 reports the performance of the five trading bots corresponding to $\theta \in \{0, 1, 2, 3, 5\}$, sorted by total return. The sample runs from December 2020 to the end of 2025, restricted

to dates for which both the ETF return and the HAR forecast are available.

The choice of December 2020 as the starting point for the trading evaluation is deliberate. First, focusing on the last five years ensures that the backtest is conducted over a recent period in which the short-term VIX futures ETF exhibits stable liquidity, trading volumes and product characteristics that are representative of current market conditions. Second, the December 2020–December 2025 window covers several distinct volatility regimes (from the post-COVID normalisation phase to subsequent episodes of market stress and repricing of interest rates and risk premia) and is therefore sufficiently rich to assess the behaviour of the strategy across heterogeneous environments while keeping the evaluation horizon fixed and directly comparable in calendar time.

Table 1: Performance of threshold-based volatility trading bots.

Threshold θ	Total return	Sharpe	Max drawdown
5.0	5.895	0.919	0.567
3.0	2.359	0.687	0.640
2.0	2.000	0.656	0.717
0.0	0.261	0.435	0.797
1.0	0.242	0.424	0.771

The bot with the highest threshold, $\theta = 5$, delivers the best performance: it multiplies initial capital by almost six over the backtest window (total return of approximately 5.9), with an annualised Sharpe ratio close to 0.92. However, this strategy also experiences a substantial maximum drawdown of about 0.57, i.e. a peak-to-trough loss of roughly 57% of capital. As the threshold is lowered, the strategy trades more frequently and becomes more exposed to noise in the spread. This results in lower total returns and, in some cases, larger drawdowns despite slightly improved diversification over time.

Figure 9 illustrates the cumulative equity curves for the three best-performing thresholds ($\theta = 5, 3, 2$). All three strategies benefit from pronounced volatility episodes, during which

the HAR forecast strongly exceeds VIX and the bot is positioned long volatility. At the same time, periods of sideways or mean-reverting volatility can produce extended drawdowns, highlighting the risk inherent in trading short-term volatility products.

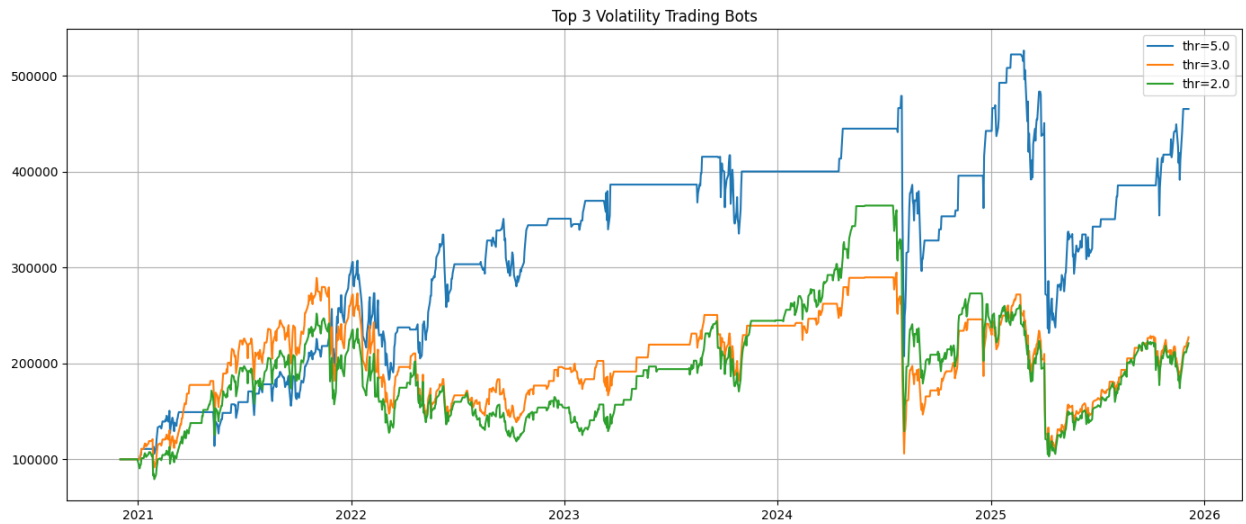

Figure 9: Cumulative equity curves for the top three volatility trading bots (thresholds $\theta = 5, 3, 2$).

Overall, this experiment confirms that the forecast–implied spread generated by the HAR model contains exploitable information for volatility trading. At the same time, the magnitude of the observed drawdowns and the sensitivity of performance to the mispricing threshold underscore the importance of risk management, trading frictions and model risk in any practical implementation of a volatility trading bot based on forecast realised volatility.

11 Conclusion

This study applies a heterogeneous autoregressive (HAR) model to realised volatility for the S&P 500 index, with a particular focus on the relative role of volatility components measured at 1-, 3- and 6-month horizons and on the practical use of volatility forecasts in a simple VIX-based trading bot.

Starting from daily index data, we construct monthly realised volatility measures at the

1-, 3- and 6-month horizons and estimate their impact on next-month realised volatility. The main empirical findings can be summarised as follows:

(i) The one-month realised volatility component is the dominant driver of next-month volatility. It accounts for the bulk of the model's explanatory power, remains strongly significant under both conventional and HAC-robust inference and contributes almost entirely to the semi-partial R^2.

(ii) The three-month component provides virtually no incremental explanatory content once the one-month term is included. Its coefficient is statistically insignificant across specifications and its semi-partial R^2 is essentially zero.

(iii) The six-month component may capture some slow-moving volatility trends, but its impact is small, statistically fragile and economically limited when assessed in terms of its marginal contribution to model fit.

(iv) Residual diagnostics reveal marked departures from normality, with strong excess kurtosis and evidence of volatility clustering. These features justify the use of HAC-robust standard errors and are in line with the stylised facts for volatility processes.

(v) Out-of-sample forecasts over 2019–2025 show that the HAR model has meaningful predictive content and is able to track major volatility episodes, including the COVID-19 shock and subsequent periods of elevated market uncertainty, while remaining competitive with simple benchmarks based on constant or historical-average volatility.

(vi) When linked to option-implied volatility from VIX, the HAR-based forecasts generate a forecast–implied spread that can be exploited in a simple threshold-based volatility trading bot. As documented in Table 1, the version with the highest mispricing threshold ($\theta = 5$) achieves the best performance, multiplying initial capital by almost six over the backtest window and delivering an annualised Sharpe ratio close to 0.9, albeit at the cost of substantial maximum drawdowns of around 55–60%. Lower thresholds lead

to more frequent trading and lower overall profitability, with limited improvement in risk-adjusted performance.

Overall, the evidence supports the view that short- to medium-term volatility, summarised by the most recent month, is the key determinant of near-term risk, while longer-horizon components play a secondary role. The HAR framework provides a parsimonious and interpretable tool for capturing these dynamics and offers a transparent bridge between time-series volatility modelling and volatility trading applications. At the same time, the trading results highlight the importance of risk management: large drawdowns and sensitivity to the choice of the mispricing threshold suggest that transaction costs, leverage constraints and dynamic position sizing would be crucial in any practical implementation of a volatility trading bot based on forecast realised volatility.

Extensions could consider richer specifications incorporating leverage effects, asymmetries with respect to negative returns, regime shifts or additional information from options markets and macro-financial variables, as well as more sophisticated portfolio and risk-management overlays for volatility-arbitrage strategies built on HAR forecasts.

References

Corsi, F. (2009). A Simple Approximate Long-Memory Model of Realized Volatility. *Journal of Financial Econometrics*, 7(2), 174–196.

Newey, W.K., and West, K.D. (1987). A Simple, Positive Semi-definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix. *Econometrica*, 55(3), 703–708.